Paul Schulte

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Education

RWTH Aachen University — Oct 2025 – Present
M.Sc. Computer Science; current grade: 1.2 (1.0 best) — Aachen, Germany
- **Focus**: Machine Learning, Optimization, Algorithms, and Graph Theory

RWTH Aachen University — Oct 2022 – Aug 2025
B.Sc. Computer Science; final grade: 1.1 with distinction (1.0 best) — Aachen, Germany
- Completed the six-semester degree in five semesters
- **Bachelor's thesis** (grade: 1.0): *Action Classification Through Dynamic Scene Graph Generation from Image Sequences*

Aalto University — Aug 2024 – Dec 2024
Semester abroad, Computer Science; grade: 5.0 / 5.0 — Espoo, Finland
- **Coursework**: Competitive Programming, Distributed Algorithms, Graph Theory

Research Experience

RWTH Aachen University, Chair for Learning on Graphs — Aachen, Germany
Graduate Research Assistant — Oct 2026 – Present
- Researching Algorithmic Foundation Models in Christopher Morris's group

RWTH Aachen University, Chair for Machine Learning and Inference — Aachen, Germany
Graduate Research Assistant — Sep 2025 – Aug 2026
- Developed a neuro-symbolic framework for learning lifted planning models directly from image sequences in Hector Geffner's group
- Designed a spatio-temporal graph neural network to infer action types and involved objects from dynamic scene graphs
- Evaluated generalization across planning domains and unseen environments with varying numbers of objects
- Co-first-authored a paper presented orally at the GenPlan Workshop at ICAPS 2026

Robotics Institute, Carnegie Mellon University — Pittsburgh, PA, USA
Undergraduate Research Intern (RISE Worldwide) — May 2024 – Sep 2024
- Developed a model predictive control framework with control barrier functions for safety-critical trajectory planning
- Implemented real-time planning in JAX for CPU-based F1TENTH hardware, reducing reliance on GPU-heavy approaches
- Tested the system on a physical F1TENTH car, successfully avoiding other cars and obstacles
- Co-authored the USDOT/Safety21 final report *Risk-Aware Warning and Control for Interactive Traffic Safety*

Professional Experience

Siemens Logistics — Düsseldorf, Germany
Software Engineering Working Student — Feb 2023 – Nov 2024
- Designed and implemented C# features for an airport baggage handling system planning tool
- Optimized computationally intensive planning components, reducing runtime by up to 80%

Publications

Ulzhalgas Rakhman*, **Paul Schulte***, Hector Geffner. *Learning Lifted Symbolic Planning Domains from Image Sequences (VISA)*. GenPlan Workshop at ICAPS 2026. Oral presentation. *Equal contribution.

Technical Skills

Programming: Python, C++, Java
ML / Scientific Computing: PyTorch, NumPy
Tools: Linux, Git, Docker, Slurm, LaTeX
Languages: German (native), English (C1)

Social Engagement

First Board Member and Spokesperson, Student Dormitory — Oct 2025 – Present
- Represent over 100 residents and liaise with the student housing administration, addressing concerns and coordinating solutions
- Chair meetings, coordinate residents and organize events that support the dormitory community

Honors & Distinctions

Dean's List: Top 5% of degree program (2022/23, 2023/24, 2024/25)
DAAD RISE Worldwide Scholarship (2024)
German Mathematical Society: One-year student membership award (2022)
German Physical Society: One-year student membership award (2022)